

AKASH KUNDU

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Portfolio | GitHub | LinkedIn

EXPERIENCE

Purposive AI | AI Research & Social Innovation Intern (Hybrid, Part-Time)

Kolkata, India | Jul 2026 - Present

Python, FastAPI, LangGraph, PostgreSQL, Sarvam AI, Ollama

- Cut manual bookkeeping effort for SHG women by **80%** by building the **Voice-Ledger** Python FastAPI pipeline, auto-extracting Bengali voice notes into bank-submittable PDFs.
- Minimized scheme-matching hallucinations to **0%** by building a **RAG PDF chatbot** with **2-stage** per-chunk grounding checks to catch mismatched citations.
- Slashed AI vendor lock-in and inference costs by eliminating OpenAI dependencies, serving **100%** of requests via a Sarvam AI to Ollama fallback cascade.
- Sustained a **100% pass rate** across a **156-test** PyTorch offline suite, preventing regressions in ledger validation and RAG grounding logic.
- Orchestrated agents via LangGraph state-machines and PostgreSQL, k-anonymizing market trends across **5 sellers** and offloading tasks to Celery within **20s** WhatsApp deadlines.

EDUCATION

Techno India University

Degree in B.Tech in Computer Science

CGPA: 7.5

Kolkata, West Bengal

2023 - 2027

Arambagh Vivekananda Academy

Degree in Higher Secondary (AISSCE)

Percentage: 71.60%

Arambagh, West Bengal

2021 - 2023

PROJECTS / OPEN-SOURCE

OphthalmolAI – AI-Based Eye Disease Predictor | Personal Project

Dec 2025 - Apr 2026

Python, FastAPI, PyTorch, OpenCV, MobileNetV3, EfficientNet-B4, Docker, Kubernetes, Linux CLI, Git

- Fine-tuned an EfficientNet CNN in PyTorch to **>85% accuracy** across **7 ocular diseases**, implementing automated unit tests covering the data-preprocessing-to-inference pipeline.
- Accelerated epoch training speed **70%** by migrating and troubleshooting the PyTorch training environment inside a GPU-accelerated Docker container.
- Streamlined clinical review time **40%** via an automated Grad-CAM PDF reporting pipeline scripted end-to-end from the Linux CLI.
- Shipped **3 production** FastAPI REST API endpoints backed by a **66-test suite**, fixing **3 critical bugs** to prevent production HTTP crashes.
- Categorized **7 eye conditions** by urgency across **3 anatomical groups**, implementing Alembic database schema migrations verified through **100% clean** upgrade/downgrade test cycles.

COPPER – AI Desktop Productivity Assistant | Personal Project

Apr 2026 - Present

Tauri, FastAPI, React, TypeScript, LangCrew, Ollama, Local LLMs, SQLite, ChromaDB, Redis

- Architected an offline agentic LLM system orchestrating **30 specialized agents** across **8 local LLM pools** via LangCrew, delivering **sub-50ms inference** with zero external API calls.
- Boosted workflow efficiency **30%** by engineering an autonomous voice-to-text agent running on consumer hardware (**6GB VRAM**) using sequential agent hot-swapping.
- Designed a **4-tier** Guardian Alignment safety framework (Levels 0-3) and a self-healing **3-stage retry** engine for destructive CLI terminal command review.
- Implemented **42 unit tests** validating the assistant fully offline and network-sandboxed, processing simultaneous daily voice queries with zero data leakage.
- Engineered a persistent **13-section** Tauri React desktop UI with push-to-talk voice I/O and secret-masked terminal output for safe command review.
- Structured long-term memory into **3 confidence classes** (Fact, Observation, Hypothesis) across **14 SQLite tables** and a ChromaDB vector store.

SKILLS

Programming Languages:

Python, C++

Frameworks / Libraries:

FastAPI, PyTorch, LangGraph, LangCrew, React, OpenCV

Tools / Platforms:

Git, Docker, GitHub Actions, Ollama, Linux CLI

Databases:

PostgreSQL, MongoDB, SQLite

CERTIFICATIONS

• Microsoft Certified: Machine Learning Operations Engineer Associate (AI-300)

Aug 2026

• Oracle AI Database Certified Foundations Associate (1Z0-1195-26)

Jul 2026

• Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103)

Jun 2026

• Oracle Agentic AI Certified Foundations Associate (1Z0-1157-26)

Jun 2026